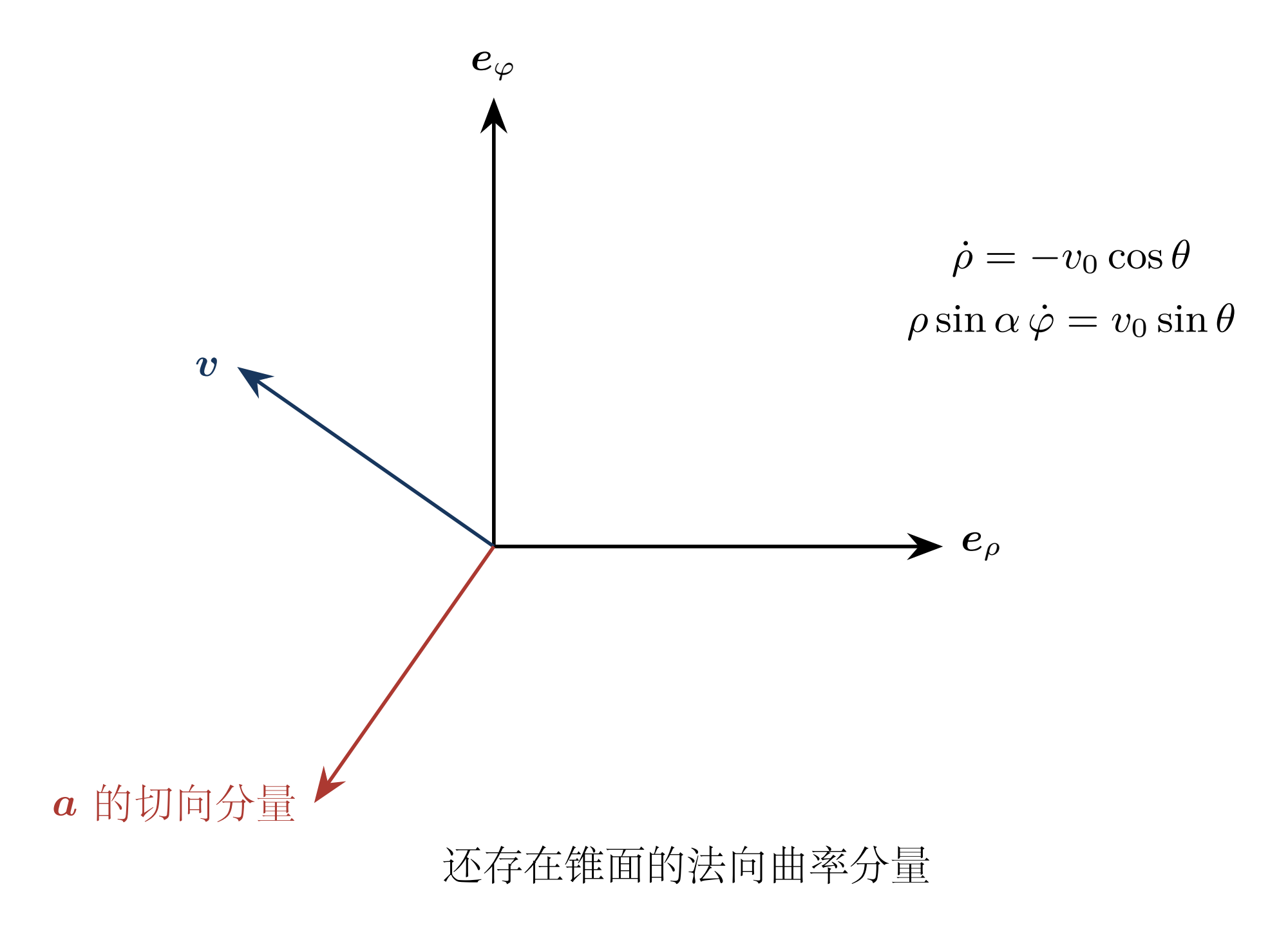


$$\dot{\rho} = -v_0 \cos\theta$$

$$\rho \sin\alpha \, \dot{\varphi} = v_0 \sin\theta$$

还存在锥面的法向曲率分量